Proposition 5: the lower bound 4/3 for the edit distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Proposition5_EditLowerBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Proposition 5 (Section 3.1, p. 6, l. 221–223; proof in Appendix C.1, pp. 21–24, l. 675–758): *For every odd $N \geq 3$, there are graphs G and H with $d_{\mathrm{ed}}(G,H) \geq 4N-2$ and $d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \leq 3N+1$. Thus, for every $\varepsilon > 0$, there is a sufficiently large N, such that $\delta_{\mathrm{ed}}(G,H) \geq (\frac{4}{3}-\varepsilon)\,\delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$.* The second sentence gives Theorem 3 (1) (p. 4, l. 124).

Lean. The graphs are $G =$ `GN N` and $H =$ `HN N` on the common vertex set `VN N` (`Construction.lean`). All statements are in the namespace `GsimOT.Prop5`.

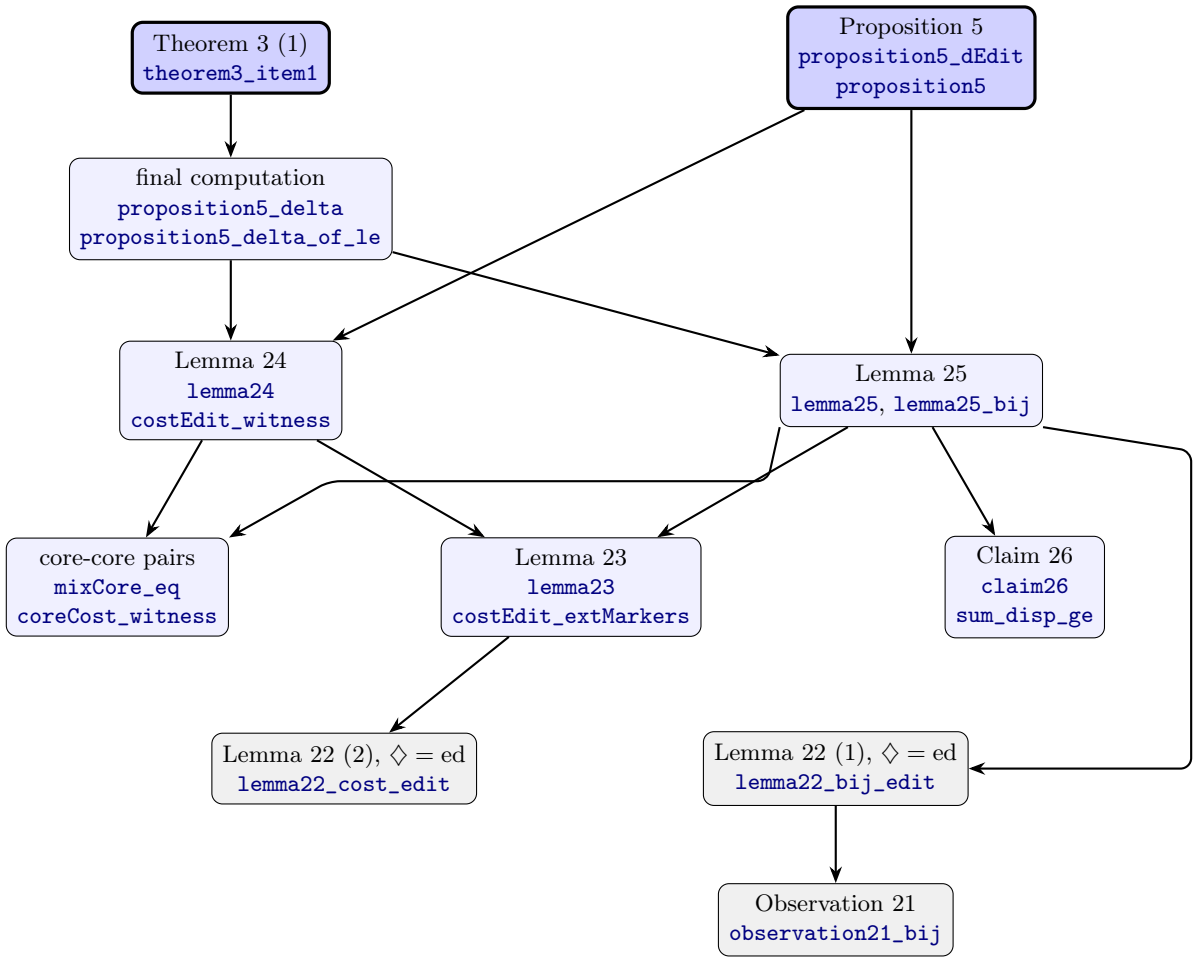
- `proposition5_dEdit`: if N is odd and $N \geq 3$, then $4N-2 \leq d_{\mathrm{ed}}(G,H)$ and $d_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \leq 3N+1$ (the paper's first sentence).
- `proposition5`: the same hypotheses give $4N-2 \leq C_{\mathrm{ed}}(\pi)$ for every bijection π and an explicit $Q \in \mathrm{DS}(V,W)$ with $C_{\mathrm{ed}}(Q) \leq 3N+1$ (the form with a witness).
- `proposition5_delta_of_le`: if N is odd, $N \geq 3$ and $\frac{10}{3(3N+1)} \leq \varepsilon$, then $(\frac{4}{3}-\varepsilon)\,\delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \leq \delta_{\mathrm{ed}}(G,H)$.
- `proposition5_delta` and `theorem3_item1` (Theorem 3 (1)): for every $\varepsilon > 0$ there is an odd $N \geq 3$ with $(\frac{4}{3}-\varepsilon)\,\delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H) \leq \delta_{\mathrm{ed}}(G,H)$.

Here C_{ed}, d_{ed}, $d_{\mathrm{ed}}^{\mathrm{OT}}$, $\delta_{\mathrm{ed}} = d_{\mathrm{ed}}/n^2$ and $\delta_{\mathrm{ed}}^{\mathrm{OT}}$ are `costEditBij`/`costEdit`, `dEdit`, `dOTEdit`, `deltaEdit`, `deltaOTEdit` of `Common/EditDistance.lean`, the same definitions as for Theorem 6 and Theorem 15. d_{ed} is an infimum over bijections and $d_{\mathrm{ed}}^{\mathrm{OT}}$ an infimum over $\mathrm{DS}(V,W)$.

2 Where the proof lives

File	Content
`Construction.lean`	G', H' (core, markers, semicircles I_r), their degrees, the marker order, and G, H by Lemma 22 with $\Delta = N+1$, $\tau = 4N-1$, $\delta = 1$ (Appendix C.1, l. 677–694).
`Lemma23_MarkerCost.lean`	the cyclic distance ℓ, compliant alignments, the cost split, Lemma 23.
`CoreEdges.lean`	counting the core-core pairs: mixCore $= 2(N-1) - 2\,$pres; averages of permutation matrices.
`Lemma24_FractionalCost.lean`	$\pi_{\pm}$, the witness $Q = \frac{1}{2}(P_{\pi_-} + P_{\pi_+})$, $C_{\mathrm{ed}}(Q) = 3N$, Lemma 24.
`Lemma25_BijectionCost.lean`	restriction to $V(G')$, displacements, Claim 26, Lemma 25.
`Proposition5.lean`	Proposition 5, its final computation, Theorem 3 (1).
`Common/Lemma22_LockingMarkers*.lean`, `Common/Observation21_DegreeBound.lean`	Lemma 22 (locking markers) and Observation 21 (shared with Proposition 8).

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`. Nothing is cited; all graphs are built in `Construction.lean`, which uses `lockGraph` and `lockSize` of Lemma 22.

4 The proof step by step

Construction (Appendix C.1, pp. 21–22, l. 677–694; Figures 1, 4). The graphs G', H' share the markers $z_0, \dots, z_{N-1}$; the cores are $S_G = \{v_0, v_2, \dots, v_{2N-2}\}$ and $S_H = \{w_1, \dots, w_{2N-1}\}$, placed at their indices on $\mathbb{Z}_{2N}$. Distinct core vertices are adjacent iff their positions sum to 0 mod $2N$, and z_r is adjacent to the core vertices on $I_r = \{(r+s) \bmod 2N \mid s < N\}$. Lean: `Base N = Fin N ⊕ Fin N` is the common vertex set of G' and H': `Sum.inl c` is v_{2c} in G' and w_{2c+1} in H' (the paper's identification $v_i \leftrightarrow w_{i+1}$), `Sum.inr r` is z_r; `posG`, `posH`, `semicircle`, `coreGraph`, `Gp`, `Hp`. For odd N the core edges are $\{c, c'\}$ with $c + c' = N$ in G' and with $c + c' + 1 = N$, $c \neq c'$ in H' (`Gp_adj_inl_inl`, `Hp_adj_inl_inl`); v_0 and w_N are isolated (l. 692–693, 722–723).

The graphs G, H are obtained by Lemma 22 with the markers ordered by their degree in G' (`markerEnum`, via `Tuple.sort`; `markerEnum_mono`) and $\Delta = N + 1$, $\tau = 4N - 1$, $\delta = 1$ (l. 688–690): `GN N = lockGraph (Gp N) (markerEnum N) (N + 1) (4 * N - 1) 1`, likewise `HN N`, and the clique has `mK N = lockSize N (N+1) (4N-1) 1` $= 2\Delta + 2\tau + k(\tau + \delta) + 1$ vertices. The paper's validity check of the parameters (l. 690–691) is `deg_Gp_le`, `deg_Hp_le` (all degrees $\leq N + 1$) and `deg_Gp_inr`, `deg_Hp_inr`, `markerEnum_deg` (the degrees of z_r differ by one); these are exactly the hypotheses of `lemma22_bij_edit`.

Compliant alignments and ℓ (l. 695–699). A compliant Q is `extendId (extMarkers R)`: R on the core vertices, extended by the identity on the markers (`extMarkers`) and on the clique (`extendId`, Common). $\ell(v_i, w_j) = \min_k |i - j + 2kN|$ is `cycDist`/`ell`.

Lemma 23 (p. 22, l. 700–706). Mapping v_i to w_j changes the adjacency to exactly $\ell(v_i, w_j)$ markers: `card_semicircle_ne` (the number of semicircles containing exactly one of two positions is their cyclic distance), summed in `sum_marker_inl`, `sum_marker_inr`. Counting both orientations, `costEdit_extMarkers`

splits C_{ed} w.r.t. (G', H') into the core-core part `coreCost N R`, the core-marker part $2\sum R(c,d)\,\ell(c,d)$, and 0 for marker pairs; `lemma23` transfers this to (G, H) by Lemma 22 (2) (`lemma22_cost_edit`).

Lemma 24 (pp. 22–23, l. 707–725). $\pi_\pm(v_i) = w_{i\pm1}$; with the identification, π_+ is the identity on the core and π_- is `predCore` ($c \mapsto c - 1 \bmod N$). $Q = \frac{1}{2}(P_{\pi_-} + P_{\pi_+})$ is `Qwit N = extendId (extMarkers (Rwit N))`, `Rwit = avgPerm`; `Qwit_isDS`. Core-marker pairs: $\ell(v_i, \pi_\pm(v_i)) = 1$ (`ell_self`, `ell_predCore`), so they give $2N$ (`sum_ell_witness`, l. 713–714). Core-core pairs (l. 715–724, Figure 5): `coreCost_avgPerm` writes them as the average of the four cases $(\pi_\pm, \pi_\pm)$, and `mixCore_eq` gives each case as $2(N-1) - 2\,\mathrm{pres}$, where `pres` counts preserved ordered core edges; `pres_id_id`, `pres_pred_pred` (no core edge preserved) and `pres_id_pred`, `pres_pred_id` ($N-2$ of the $N-1$ ordered core edges preserved, the exceptional one of Figure 5b is lost) give $\frac{1}{4}(2(N-1) + 2 + 2 + 2(N-1)) = N$ (`coreCost_witness`). Hence $C_{\mathrm{ed}}(Q) = 3N$ (`costEdit_witness`) and $d_{\mathrm{ed}}^{\mathrm{OT}} \le 3N \le 3N+1$ (`lemma24`, via `dOTEdit_le`).

Lemma 25 (pp. 23–24, l. 726–753). Let π be a bijection. If $C_{\mathrm{ed}}(\pi) \ge \tau = 4N-1$, done (the case split in `lemma25_bij`). Otherwise Lemma 22 (1) (`lemma22_bij_edit`) gives that π fixes the markers and maps $V(G')$ onto itself (l. 727–728). The paper then replaces π on the clique by the identity (l. 728–729); Lean instead restricts π to $V(G')$ (`restrictInl`, applied twice to get the core map ρ) and drops the pairs that involve the clique, which only lowers the cost (`costEditBij_restrict`). The remaining cost is `mixCore N rho rho` plus $2\sum_c \ell(v_{2c}, \pi(v_{2c}))$ (`costEditBij_sumCongr`, by Lemma 23). Displacements $d_i \in \{-N+2, \dots, N\}$ (l. 729–732) are `disp`; $\ell(v_i, \pi(v_i)) = |d_i|$ (`ell_eq_abs_disp`), $|d_i| \ge 1$ (`one_le_ell`); $U = \sum |d_i|$ and $2k =$ `pres N rho rho`.

Claim 26 (p. 24, l. 737–749), `claim26`: $U \ge N + 2k$. The case $d_i = N$ (l. 738–739) and the general case follow the paper: $\sum d_i \equiv N \pmod{2N}$ (`sum_disp`, l. 740–741), hence $|\sum d_i| \ge N$ (`sum_disp_ge`, l. 742); a preserved edge has $d_i + d_{i'} = 0$ (`Preserved`, `disp_add_disp_partner`, l. 743–745); the set S of endpoints of preserved edges has $|S| = 2k$ (`pres_eq_card`) and $\sum_S d_i = 0$ is proved with the involution $c \mapsto N - c$ on S (l. 746–749).

Finally (l. 750–753) the core-core pairs cost $2(N-1-2k)$ (`mixCore_eq`, `sum_adj_Gp_core`, `sum_adj_Hp_core`) and $C_{\mathrm{ed}}(\pi) \ge 2(N-1-2k) + 2U \ge 4N-2$ (`lemma25_bij`); `lemma25` takes the infimum (`le_dEdit`).

Proof of Proposition 5 (p. 24, l. 754–758). `proposition5_delta_of_le` is the paper's chain $d_{\mathrm{ed}} \ge 4N-2 = (\frac{4}{3} - \frac{10}{3(3N+1)})(3N+1) \ge (\frac{4}{3} - \frac{10}{3(3N+1)})\, d_{\mathrm{ed}}^{\mathrm{OT}} \ge (\frac{4}{3} - \varepsilon)\, d_{\mathrm{ed}}^{\mathrm{OT}}$ (a `calc` block, using `lemma24`, `lemma25` and $\frac{4}{3} - \frac{10}{3(3N+1)} \ge 0$), followed by division by the common normalisation n^2 (l. 757). `proposition5_delta` picks the odd $N = 2k+3$ with $k > \frac{10}{9\varepsilon}$, so that $\frac{10}{3(3N+1)} \le \varepsilon$ (l. 757–758).

5 Assumptions and trusted base

No cited result and no hypothesis beyond N odd, $N \ge 3$ (and $\varepsilon > 0$). The main theorems depend only on `propext`, `Classical.choice`, `Quot.sound` (checked by `scripts/CheckAxioms.lean`).

6 Notes on the formalisation

- **Statement.** `proposition5_dEdit` is the paper's statement; `proposition5` is the form with an explicit witness Q. The "sufficiently large N" of the second sentence is made explicit by `proposition5_delta_of_le` (every odd $N \ge 3$ with $\frac{10}{3(3N+1)} \le \varepsilon$).
- **Lemma 24.** The Lean computes the cost of the witness, $C_{\mathrm{ed}}(Q) = 3N$ (`costEdit_witness`), and `lemma24` derives the bound $3N+1$ from it.
- **Lemma 25.** Dropping the clique pairs instead of replacing π on the clique by the identity (same effect: the cost only decreases). Claim 26 is stated with $2k =$ `pres N rho rho`, the number of preserved *ordered* core edges.
- **Vertex sets.** G and H live on the same type `VN N`; the identification $v_i \leftrightarrow w_{i+1}$ is built into `Base N`.
- **Lemma 22 for $\diamondsuit$ = ed** (`Common/Lemma22_LockingMarkersEdit.lean`): item (1) is derived from the case $\diamondsuit = 1$ via $C_1(\pi) \le C_{\mathrm{ed}}(\pi)$ (`costOneBij_le_costEditBij`); the paper proves both cases at once from Eq. (16) and Observation 21.

7 How to check this result

- Build: `lake build GsimOT.Proposition5_EditLowerBound.Proposition5` (or `bash build.sh` for the whole library, which also runs `scripts/CheckAxioms.lean`).
- Axioms: `#print axioms GsimOT.Prop5.proposition5_dEdit`, and likewise for `proposition5`, `proposition5_delta` `proposition5_delta_of_le`, `theorem3_item1`; expected output: `propext`, `Classical.choice`, `Quot.sound`.
- The table `Proposition5_correspondence.pdf` in this folder lists every paper item with its Lean declaration and source location.